claude-windows / fa4dc4eb-4705-47fd-b044-4f017b586b15

Metadata

- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\fa4dc4eb-4705-47fd-b044-4f017b586b15\subagents\workflows\wf_ac0cbb29-6bf\agent-aee626016177f5958.jsonl`
- SHA-256: `331b0ebf94455cdfbd9681e17848759c45795f52a67ba6ca67551a2a4da2e679`
- Source modified: `2026-06-19T15:49:15+00:00`
- Imported at: `2026-07-05T16:48:22+00:00`
- project: `wf_ac0cbb29-6bf`
- session_id: `fa4dc4eb-4705-47fd-b044-4f017b586b15`

Transcript

1. user (2026-06-19T15:46:34.350Z)

You are auditing a badminton rally-detection codebase for DOC/COMMENT STALENESS ? statements that CONTRADICT the current code or data.
HARD RULES:
- Report ONLY factual staleness with CODE EVIDENCE (file:line or symbol) that proves it. Open the cited code and confirm before reporting.
- NEVER invent a fact or a status. If a thing is built-but-default-OFF / unwired / unmeasured, say EXACTLY that ? do not overclaim "in production".
- A "[design]" tag is stale ONLY if the thing is actually BUILT in code (cite the symbol).
- DO NOT propose subjective rewrites, style/tone changes, or improvements. DO NOT relitigate owner-gated DECISIONS (owned-model roadmap, licensing/MIT-Apache stance, product/strategy choices, genuinely-still-design proposals).
- Preserve the document's voice and formatting; make the SMALLEST surgical edit. stale_text MUST be copied verbatim with enough context to be unique.
- IGNORE docs/archives/** (terminal snapshots) and any memory/ file.
Working dir is the repo root; use Read/Grep/Glob freely.

YOUR SLICE ? meta:
Top-level meta staleness vs code: CLAUDE.md, docs/README.md, docs/NEXT_STEPS.md. SEED 1 (verify): CLAUDE.md says "This dev container lacks GPU / torch / cv2 / numpy ? the full pipeline can not run here" ? STALE: the repo runs the full WSL/GPU/WASB pipeline and cv2/numpy/torch import fine here. SEED 2 (verify against backend/config/models.py): the cap=6 + R4 inactivity-end milestone is now DEFAULT-ON (max_window_duration=6.0, window_inactivity_end=1.5, inactivity_rally_thresh=0.20, inactivity_min_density=0.30) ? flag any meta doc still calling local windowing not-started / default-OFF / not-flipped / scaffolding-only. Only fix factual contradictions, not roadmap intent.

Return EVERY code-contradicting staleness item you can verify in your slice. Exhaustive within the slice; facts only.

2. assistant (2026-06-19T15:46:37.047Z)

I'll audit the three meta docs in my slice for staleness against the current code. Let me start by reading the key files and verifying the seeds.

3. user (2026-06-19T15:46:38.062Z)

```
"""Pydantic models for the application configuration.

This module defines the canonical, validated, typed configuration for the
Badminton Highlight Indexer. Built-in defaults live here (single source of truth).

The models support graceful migration from old config.json files via extra="allow".
"""

```

```python
from __future__ import annotations

from typing import Any, List, Literal, Optional

from pydantic import BaseModel, Field, field_validator, model_validator

Sport = Literal["badminton", "tennis", "pickleball", "table_tennis", "cricket"]


class ValidationRelevanceConfig(BaseModel):
    court_green_lower: list[int] = Field(default=[35, 40, 40])
    court_green_upper: list[int] = Field(default=[85, 255, 255])
    court_pixels_min_ratio: float = Field(default=0.05, ge=0.0, le=1.0)


class ValidationConfig(BaseModel):
    min_resolution_width: int = 1280
    min_resolution_height: int = 720
    min_fps: float = 29.9
    min_blur_threshold: float = 20.0
    max_scene_cuts_per_minute: float = 0.5
    relevance: ValidationRelevanceConfig = Field(default_factory=ValidationRelevanceConfig)


class TrackNetConfig(BaseModel):
    wsl_distro: str = "Ubuntu"
    repo_dir: str = "~/models/TrackNetV4"
    conda_sh: str = "~/miniconda3/etc/profile.d/conda.sh"
    conda_env: str = "TrackNetV4"
    weights_path: str = ""
    queue_length: int = 5
    stage_in_wsl: bool = True
    wsl_stage_dir: str = "~/clips"
    timeout_sec: int = 1800  # 30 min; kills a hung WSL/GPU call (0 = no timeout)
    velocity_threshold: float = 0.02
    # Cache hygiene: after a SUCCESSFUL run the staged video copy (and, for WASB, the
    # decoded frame cache) are deleted automatically ? they are pure intermediates,
    # regenerable from the source, and the dominant disk consumer (~GBs/video). Set
    # true only for debugging. On FAILURE they are always kept so a re-run can resume.
    keep_frames: bool = False
    # Warn before a run if WSL free space (at wsl_stage_dir) is below this many GB.
    # 0 = disabled.
    min_free_gb: float = 0.0


class WasbIndexConfig(BaseModel):
    """Typed config for the live WASB shuttle substrate (indexing.wasb).

    Mirrors backend/pipeline/detectors/wasb_runner.py::WasbConfig.from_indexing_cfg so
    these knobs actually take effect ? previously this whole block was silently dropped
    because nested config sections defaulted to extra='ignore'.
    """
    wsl_distro: str = "Ubuntu"
    repo_dir: str = "~/models/WASB-SBDT"
    conda_sh: str = "~/miniconda3/etc/profile.d/conda.sh"
    conda_env: str = "wasb"
    weights_path: str = "~/models/WASB-SBDT/pretrained_weights/wasb_badminton_best.pth.tar"
    sport: str = "badminton"
    wsl_stage_dir: str = "~/clips"
    timeout_sec: int = 1800  # 30 min; kills a hung WSL/GPU call (0 = no timeout)
    velocity_threshold: float = 0.02
    # Cache hygiene: after a SUCCESSFUL run the staged video copy + decoded frame cache
    # (the ~GBs/video disk hog) are deleted automatically ? pure intermediates,
```

```
regenerable
    # from the source. Set true only for debugging. On FAILURE they are kept so a re-run resumes.
    keep_frames: bool = False
    # Warn before a run if WSL free space (at wsl_stage_dir) is below this many GB. 0 = disabled.
    min_free_gb: float = 0.0
    # FAST PATH (default OFF until owner-verified side-by-side): decode the staged video
    # on the fly and feed frames straight to the detector, skipping the slow per-frame PNG
    # extraction that dominates a long clip (~46k PNGs for a 12-min 1080p60 video, which
    # overran the 30-min timeout). Output is bit-identical to the PNG path (PNG is lossless),
    # so this is purely a speed/disk win; kept opt-in until the side-by-side confirms it.
    stream_video: bool = False


class FusionConfig(BaseModel):
    """Config for the multi-signal `fusion_hybrid` segmenter (MULTI_SIGNAL_FUSION_PLAN P2).

    Every default reproduces control behaviour: the fusion segmenter persists the
    shuttle net-crossing count (Gate A signal) and ? only when ``cue_flags['person']`` is
    true ? the person-cue features, but it does NOT gate the served handoff unless a
    threshold is raised (``min_crossings``/``min_players`` default 0 = OFF). The court mask
    is optional: ``court_polygon=None`` ? Gate B counts every detection (player-count-only);
    supplied ? off-court persons (spectators / adjacent courts) are excluded.
    """
    # Which trajectory substrate seeds the windows (shuttle stays primary).
    substrate: Literal["wasb", "tracknet"] = "wasb"
    # Windowing velocity threshold for the fusion family (the engine reads
    # indexing.<family>.velocity_threshold). Default matches the substrates' 0.02; set
    # equal to indexing.wasb.velocity_threshold to A/B fusion against a tuned wasb run.
    velocity_threshold: float = 0.02
    # Per-cue enable flags, e.g. {"person": true}. Person cue is OFF unless explicitly
    # enabled ? so the default path never imports torch / touches the GPU.
    cue_flags: dict[str, bool] = Field(default_factory=dict)
    # Served Gate A: drop windows whose shuttle net-crossings < this from the AI handoff.
    # 0 = OFF (no gating; persisted candidates are always the full superset for offline re-scoring).
    min_crossings: int = Field(default=0, ge=0)
    # Served Gate B: when the person cue is on, drop windows with median in-court players
    # < this. 0 = OFF.
    min_players: int = Field(default=0, ge=0)
    # Optional court mask as normalised 0-1 [[x, y], ...] polygon. None = no mask.
    court_polygon: Optional[list[list[float]]] = None
    # Net line for the person two-sided-motion split (person features only ? the shuttle
    # net-crossing gate keeps rally_gate's camera-agnostic auto-estimate).
    net_axis: Literal["x", "y"] = "y"
    net_position: float = Field(default=0.5, ge=0.0, le=1.0)


class IndexingConfig(BaseModel):
    default_segmenter: str = "yolo_hybrid"
    use_gpu: bool = True
    motion_threshold: float = 0.08
    min_segment_duration: float = 3.0
    dead_time_merge_gap: float = 2.0
    chunk_duration: float = 90.0
```

```
        chunk_overlap: float = 10.0
        detector_model: str = "fasterrcnn_resnet50_fpn"
        detector_score_threshold: float = 0.5
        yolo_velocity_threshold: float = 0.15
        yolo_frame_skip: int = 3
        yolo_tracker: str = "bytetrack.yaml"
        yolo_prefetch_workers: int = 2
        min_action_window_duration: float = 2.0
        # BOUNDED WINDOWING (over-merge fix W1, RALLY_QUALITY_RESEARCH.md §6). 0 = OFF; > 0 = a window
        # longer than this many seconds is recursively split at its largest internal inactivity gap (?
        # ``window_min_split_gap`` s ? the most likely rally boundary), so one window can't swallow
        # several rallies + the dead time between them. Never merges, only cuts (recall can't drop).
        # *** R1 MILESTONE DEFAULT-ON (cap=6): the smallest safe local-windowing flip. Measured n=13
        # golden, R2 tolF1: baseline?milestone (cap=6 + R4 below) ?+0.222, sign 12/0, p=0.0005; cap=6
        # beats cap=12 ?+0.110, sign 12/0; net over-seg guard PASS (mean merge_rate 0.651?0.161). ***
        max_window_duration: float = 6.0
        window_min_split_gap: float = 0.5  # min internal gap (s) that qualifies as a split point
        # HARD-CAP fallback for W1: when True, an over-long window with NO internal inactivity gap
        # (a continuously-active trajectory ? e.g. the real-footage mahadevpura-1 174s blob) is
        # chopped at its midpoint until ? max_window_duration, making the cap a true upper bound.
        # Only cuts (recall can't drop). OFF by default until measured/promoted.
        window_hard_cap: bool = False
        # R4 rally-END inactivity trim (s, 0 = OFF). After a rally the shuttle keeps moving slowly
        # (picked up / walked) above velocity_thresh, so the window over-extends its END (the measured
        # |?end| gap vs golden). Trim the post-rally tail back to the last frame whose trailing
        # window stays dense with FAST motion (velocity > inactivity_rally_thresh). Only cuts.
        # *** R1 MILESTONE DEFAULT-ON (1.5/0.20/0.30): on top of cap=6, R4 adds a small but significant
        # end-precision gain ? ?tolF1 +0.040, sign 9/1/3, p=0.022, |?end| 4.96?3.31s (n=13). The W1 cap
        # is the dominant lever; R4 is the marginal increment. See QUALITY_ITERATIONS.md "R1". ***
        window_inactivity_end: float = 1.5
        inactivity_rally_thresh: float = 0.20
        inactivity_min_density: float = 0.30
        # R5 blob-fallback (default OFF). LAST-RESORT splitter for the continuously-active blob: after
        # the W1 gap-split AND the R4 inactivity trim have each had their chance, a group STILL longer
        # than window_blob_factor × max_window_duration (no internal gap, no rally-end signal ? e.g.
        # mahadevpura-1's ~65s residual ? 11× a 6s cap) is midpoint-chopped to ? the cap as a FORCED cut
        # (logged + flagged on the window) so the degenerate single-window outcome is avoided and the
        # clip is surfaced as needing rally-STATE cues, not a bigger cap. Differs from window_hard_cap
        # (which chops BEFORE R4 and over-segments normal clips): blob-fallback runs AFTER R4 and only
        # force-cuts the genuine blob (the factor gate). Only cuts (recall can't drop).
```

```
    window_blob_fallback: bool = False
    # Magnitude gate that keeps R5 surgical: only a group longer than this × max_window_duration is a
    # blob (a normal long rally ~1-2× the cap is left alone; the mahadevpura-1 residual is ~11×).
    # Default 4.0 is do-no-harm on the golden corpus (rescues only the continuously-active
    # clips, every other clip a no-op within noise) ? see docs/QUALITY_ITERATIONS.md "R5".
    window_blob_factor: float = 4.0
    log_yolo_candidates: bool = True
    store_yolo_candidates: bool = True
    ai_handoff_padding: float = 2.0
    ai_max_workers: int = 5
    # Width (px) the gemini segmenter re-encodes chunks to before upload. Stream-copied
    # chunks of a high-bitrate (4K) source blow past the provider upload cap
    # (backend/providers/gemini.py::MAX_UPLOAD_MB) and are refused ? observed live as
    # "0 segments / success". Re-encoding also makes chunk boundaries frame-accurate
    # (stream copy cuts at the nearest keyframe). 0 = legacy stream-copy at source res.
    # Matches gemini_refine's --clip-scale-width default.
    ai_chunk_scale_width: int = 1280
    # FULLY-LOCAL mode (default OFF): when true, the trajectory-hybrid segmenters
    # (wasb_hybrid / tracknet_hybrid / fusion_hybrid) skip the Phase-3 AI handoff entirely and
    # emit their local candidate windows AS the rallies ? no provider, no API key, nothing
    # uploaded. Lets the local detector run standalone (e.g. to benchmark it against the Gemini
    # path, or to operate with zero cloud dependency). The pure `gemini` segmenter ignores this.
    skip_ai_handoff: bool = False
    # Optional debug knob: trim every video to the first N seconds before processing.
    debug_duration: Optional[float] = None

    tracknet: TrackNetConfig = Field(default_factory=TrackNetConfig)
    wasb: WasbIndexConfig = Field(default_factory=WasbIndexConfig)
    fusion: FusionConfig = Field(default_factory=FusionConfig)

    @field_validator("default_segmenter")
    @classmethod
    def segmenter_must_be_registered(cls, v: str) -> str:
        # Registry check happens at runtime in main/segmenter selection.
        # Here we just allow any string for forward compatibility.
        return v

    @model_validator(mode="after")
    def _chunk_step_must_be_positive(self) -> "IndexingConfig":
        # The chunked segmenters advance by (chunk_duration - chunk_overlap); a non-positive
        # step makes `while t < duration: t += step` loop forever, growing memory before any
        # GPU work. Reject the bad config at load time instead.
        if self.chunk_overlap >= self.chunk_duration:
            raise ValueError(
                f"indexing.chunk_overlap ({self.chunk_overlap}) must be < chunk_duration "
                f"({self.chunk_duration}); otherwise chunking never advances."
            )
        return self


class OutputConfig(BaseModel):
    default_highlights_name: str = "highlights.mp4"
    db_name: str = "sports_indexer.db"
    # Directory where the DB, highlight reels, and processed clips are written.
```

```python
    # The CLI's --output-dir overrides this at startup (see backend/main.py::main).
    output_dir: str = "output"


class WatermarkConfig(BaseModel):
    """Branding overlay burned into each highlight clip during the per-clip re-encode
    (backend/pipeline/stitcher/compiler.py). **On by default** ? set `enabled: false`
    (under `stitching.watermark` in config.json) to turn it off. The text is fully
    configurable. The concat stage stays stream-copy (-c copy)."""
    enabled: bool = True
    text: str = "Generated by Khelsutra.guru"
    logo_path: str = ""        # optional transparent PNG overlay; "" = no logo
    font_path: str = ""        # optional .ttf; "" = use the fontconfig `font` family below
    font: str = "Sans"         # fontconfig family used when font_path is empty
    font_size_ratio: float = Field(default=0.035, gt=0.0, le=0.5)  # text height as a fraction of frame height
    opacity: float = Field(default=0.85, ge=0.0, le=1.0)
    box: bool = True           # faint backing box behind the text for legibility
    margin: int = Field(default=24, ge=0)  # px inset from the chosen corner
    position: Literal["bottom-right", "bottom-left", "top-right", "top-left"] = "bottom-right"


class StitchingConfig(BaseModel):
    begin_padding: float = 0.5
    end_padding: float = 0.5
    use_cache: bool = False
    min_shot_count: int = 1
    # Long-rally reel filter (seconds; 0 = OFF). Drop detected rally pairs whose duration
    # (end ? start) is below this from the highlight reel, BEFORE compiling. A cleaner
    # reel-selection knob than `min_shot_count`: shot_count is unlabeled / "Unknown" on the
    # fully-local no-AI path, whereas duration is derived from the rally's own start/end. The
    # local detector over-fires and its false positives are mostly SHORT (motion blips,
    # background-court fragments), so this keeps the reel to the eye-catching long rallies.
    # OFF by default ? the detector output is unchanged, only which rallies reach the reel.
    min_rally_duration: float = Field(default=0.0, ge=0.0)
    watermark: WatermarkConfig = Field(default_factory=WatermarkConfig)


class LoggingConfig(BaseModel):
    """Logging knobs for the run entrypoints (see backend/utils/logging_setup.py)."""
    # Third-party HTTP/RPC client loggers (httpx, httpcore, urllib3, google-genai,
    # google.auth, grpc) emit one INFO line per request ? dozens per Gemini run, which
    # buries our own [rally]/[compile] lines in run.log during manual QA. Default raises
    # them to WARNING; set true to restore full chatter for request-flow debugging.
    verbose_http: bool = False


class SecurityConfig(BaseModel):
    # When set, /api/process video_path must resolve under this directory (hosted/tenant
    # mode). Default None preserves the single-user local 'any local file' workflow.
    allowed_video_root: Optional[str] = None

    # --- Chunked upload (B2, PR-2) ---
    # Landing zone for browser uploads (assembled from parts). Per-user subdirectories
    # arrive with PR-3 identity scoping. ? In hosted mode, set allowed_video_root to
    # contain upload_dir or /api/process will reject the uploaded paths.
    upload_dir: str = "output/uploads"
    # Whole-file cap (default 4 GB) and per-part cap. Parts stay under ~100 MB because
```

```python
        # free-tier edge proxies cap request bodies there (HOSTING_PLAN §2).
        max_upload_mb: int = 4096
        max_part_mb: int = 95
        allowed_upload_extensions: List[str] = ["mp4", "mov"]


    class AppConfig(BaseModel):
        """Root configuration object.

        All runtime configuration should be accessed via an instance of this class
        (or the loader) rather than raw dict lookups.
        """

        sport: Sport = "badminton"
        ai_provider: str = "gemini"
        # Auto-tracking alias: 2.5-flash is legacy now that 3.5-flash is the stable flagship;
        # the alias never deprecates underneath us. Eval reports record the run date alongside.
        ai_model: str = "gemini-flash-latest"

        validation: ValidationConfig = Field(default_factory=ValidationConfig)
        indexing: IndexingConfig = Field(default_factory=IndexingConfig)
        output: OutputConfig = Field(default_factory=OutputConfig)
        stitching: StitchingConfig = Field(default_factory=StitchingConfig)
        security: SecurityConfig = Field(default_factory=SecurityConfig)
        logging: LoggingConfig = Field(default_factory=LoggingConfig)

        # Allow extra keys during transition so existing config.json files do not break.
        model_config = {
            "extra": "allow",
            "validate_assignment": True,
        }

        # Temporary migration shim for legacy dict-style access (config.get("section", {}).get(...)).
        # Will be removed once all call sites (validators, segmenters, etc.) move to typed
        # attribute access / the typed config. A nested *section* is returned as a plain dict
        # via model_dump, so legacy chained access keeps working ? returning the Pydantic
        # section model instead breaks it (a model has no .get), which bit every validator on
        # real runs. Scalar leaves are returned as-is.
        def get(self, key: str, default: Any = None) -> Any:
            _MISSING = object()
            val: Any = _MISSING
            if key in type(self).model_fields or hasattr(self, key):
                val = getattr(self, key, _MISSING)
            if val is _MISSING:
                # Legacy convenience: find a leaf key inside any nested section. Sections are
                # discovered dynamically from the model fields, so a future top-level section
                # (e.g. `audio`, `jobs`) is covered automatically without editing a hardcoded list.
                for fname in type(self).model_fields:
                    section = getattr(self, fname, None)
                    if isinstance(section, BaseModel) and hasattr(section, key):
                        val = getattr(section, key)
                        break
            if val is _MISSING:
                return default
            if isinstance(val, BaseModel):
                return val.model_dump(mode="json")
            return val
```

```
    def __getitem__(self, key: str) -> Any:
        val = self.get(key)
        if val is None and key not in type(self).model_fields:
            # Re-raise like a real dict for unknown top-level
            raise KeyError(key)
        return val


# Convenience type alias for callers that want to annotate
Config = AppConfig

```

4. user (2026-06-19T15:46:38.513Z)

```
# CLAUDE.md ? project memory / agent handoff

Local, AI-powered **badminton (? racket-sports) highlight indexer + rally detector**:
FastAPI + SQLite + ffmpeg + GPU CV/AI, with paid **Gemini** as the cloud AI. Sibling
repos: `sports-data-collector`, a VLC-based `rally-annotator`.

## Read these first
- **`docs/README.md`** ? doc index. **`docs/CODE_MAP.md`** ? code navigation + data
  contracts (the cold-start map; read before touching code).
- **Authoritative owned-model roadmap:** `docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md` (+ the
  live `docs/NEXT_STEPS.md`). These SUPERSEDE the 2026-06-07 strategy docs below where they
  disagree (licensing, Gemma-4 stance, base model, conversational-QA).
- **Strategy (post-scaffolding):** `docs/COMMERCIALIZATION.md`, `docs/PMF_MARKET_RESEARCH.md`,
  `docs/PLATFORM_ARCHITECTURE.md`, `docs/DEFERRED_HARDENING_PLAN.md`. Also `docs/ROADMAP.md`
  (offline-segmenter narrative) and `docs/BACKLOG.md`.
- **Rally-detection quality track (design merged 2026-06-13, owner-gated, NOT started):**
  `docs/HOW_RALLY_DETECTION_WORKS.md` (2-layer mental model) ? `docs/MULTI_SIGNAL_FUSION_PLAN.md`
  (fuse shuttle + person + audio cues; the held-shuttle bug is the already-built `rally_gate.py`
  "Gate A" **not wired into production** ? cheapest win) ? `docs/EXPERIMENT_HARNESS.md` (A/B + shadow
  eval so cues land flag-gated/default-OFF with zero regression) ? `docs/ROORA_LABELING_GUIDELINES.md`
  (v8 vendor labeling spec). **`docs/NEXT_STEPS.md` has the prioritized pick-up for this track.**
- **History:** `docs/archives/` ? ADRs (`decisions/DECISIONS.md`), research, and
  **`checkpoints/`** (latest: `2026-06-strategy-foundation/`). Archive policy: only move
  **terminal-state** (DONE/REJECTED) work; live docs stay at `docs/` top level.

## Current state (June 2026)
- **Scaffolding complete**; strategy foundation + owned-model roadmap merged. **No
  platform/owned-model implementation has started.** A 2026-06-10 audit-driven hardening
  sweep landed (licensing gate, detector keying, re-ingest safety, config, eval gate, API
  validation, reliability, CI + test coverage) ? see git history.
- **Two tracks, both owner-gated:** the **committed next PLATFORM slice = async job model**
  (`DEFERRED_HARDENING_PLAN.md` #16, single-tenant); the **owned-model track** starts at a
  greenlit Phase-0 milestone (`OWNED_MODEL_IMPLEMENTATION_PLAN.md`). **Do not start
  implementation without explicit owner approval.**

## Architecture in one breath
- **Pluggable registries** (ABC + `Registry` singleton): segmenters / providers /
```

validators
  / sports / annotations ? see `backend/pipeline/segmenters/base.py`. *Future:*
  `StorageBackend` + `ComputeBackend` registries (PLATFORM §3).
- `video_id` = MD5 of the file ? `backend/utils/hashing.py::compute_video_id` (one helper).
- Typed config = `backend/config/` (Pydantic). DB schema evolves via the `PRAGMA
  user_version` migration ladder in `backend/infrastructure/database.py`.

Committed guardrails (non-negotiable)
- **Paid LLMs (Gemini/OpenAI/Anthropic) = inference / serving / fallback ONLY ? never a
  training data source** (terms/copyright). Commercial-clean licensing for EVERY dependency ?
  code, data, AND weights: **MIT / Apache-2.0 / BSD only** (ratified 2026-06-09, owner-agreed).
- **Owned-model base:** **Qwen3-VL (Apache-2.0)** primary, InternVL3 (MIT) fallback; **Gemma-4
  is AMBER / bench-only** (Apache grant likely but Prohibited-Use posture unconfirmed ? counsel
  needed; do NOT build on it without sign-off). **Reject Gemma 3** (viral license). **Exclude
  minors** from the training pool; per-user training data needs a separate explicit opt-in.
- **v1 = rally + highlight + history + correction loop**; **gait/biometrics strictly future**
  (GDPR Art. 9).
- **Generic data schema**; `coach ? athlete` is an **optional overlay**, not baked in.

Workflow conventions
- **Branch off `master`; ONE PR per work item; NO PR stacking.** Don't open a PR unless asked.
- Tests: `python run_all_tests.py` (offline, fully mocked). Python 3.8+; `ffmpeg` required.
- Never push to the default branch directly; descriptive commit messages.
- **Substantial design/project work gets a tracked project doc** following `docs/PROJECT_DOC_TEMPLATE.md` (status header · decisions-locked · progress tracker · definition-of-done; archived on completion). Precedents: `docs/RALLY_QUALITY_RESEARCH.md` §7, `docs/GOLDEN_REGRESSION_FIXTURES.md`.
- ?? This dev container lacks GPU / torch / cv2 / numpy ? the **full pipeline can't run here**.
  Verify pure-logic paths and hand machine-side checks (real video, GPU/WSL suite) to the owner.

5. user (2026-06-19T15:46:38.957Z)

Next steps (live) ? owned-model + multisport

Updated: 2026-06-13 (**rally-detection quality track**: fusion + experiment-harness design docs merged,
owner-gated ? see the dedicated section below, the desktop-agent pick-up). Prior: 2026-06-10 (audit?M0?Gemini-battery session; later same day the **UI-alpha track opened**:
ADR-009 ratified ? KhelSutra Guru / khelsutra.guru, private `khelsutra` repo, local-first delivery ? see
`docs/UI_ALPHA_EXECUTION_PLAN.md`; **PR-1 async jobs (#16) MERGED 2026-06-11, PR #87**; **PR-2
upload/download MERGED 2026-06-11, PR #99** ? live-E2E-verified (3-part chunked upload MD5 byte-identity +
compile?browser-download); next: PR-3 identity. Same day: the **8?9 quality sweep #90?#98 landed** ? ruff/mypy/
coverage-floor CI lanes, SQLite conn-close fix, config unknown-key warning, hybrid-twin collapse into
`trajectory_hybrid.py`, WSL tilde+abspath live-path fixes; suite 388 green; LOVO 0.626 re-verified exact).
Authoritative roadmap:
`docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md` (+ ADR-008 for topology). **Latest blow-by-

blow:** memory
`current-state.md` / `session-handoff.md`. This file = the prioritized "what to do next," refreshed each session.

? Next-session / GPU-box pickup (2026-06-15)
M0 in-container quick wins are COMPLETE: ship-gate provisional blocker (**#170**), train/eval split guard (**#171**), the **doc-consistency sweep** (**#174** ? corrected 16 drifted docs vs the code), and the **P3 person-feature fusion litmus machinery is already present + suite-green** (`backend/eval/fusion_compare.py` + `ablation.py`: person-driven ?F1 / bootstrap CI / paired sign test; the real-golden `skipif` litmus waits only on GPU-extracted data). ? CI is red repo-wide due to a **GitHub Actions minutes/billing outage** (account-level, not code) ? work is local-gated + `--admin`-merged until Actions is restored (Settings ? Billing ? Actions minutes / spending limit).

Immediate next step ? the real P3 LOVO measurement, split GPU?CPU (the dev box is CPU-only: `torch+cpu`/`cv2` present but no GPU, and the corpus/videos/CSVs/labels are not here):
1. **[owner / GPU box]** `python -m backend.eval.fusion_golden --manifest output/human_lovo_manifest.json --out-dir "%USERPROFILE%\OneDrive\rally-annotator\golden-data\features\2026-06-15-n6"` ? produces `*_golden_features.json` for the 6 golden videos. (GPU + WSL + videos + trajectory CSVs + `.rallies.csv` labels are owner-side only.)
2. Artifacts land in the **shared cloud folder** `C:\Users\avidu\OneDrive\rally-annotator\golden-data\` (OneDrive auto-syncs both ways). Convention + workflow: **[`GOLDEN_DATA_SHARING.md`](GOLDEN_DATA_SHARING.md)**; tracked in **#175**.
3. **[CPU dev/agent session]** `python -m backend.eval.fusion_compare --features-dir <shared>\features\2026-06-15-n6 --record` + `python -m backend.eval.ablation --features-dir <shared>\... --granularity group` ? real person-feature ?F1; copy the JSONs into `output/` to also pass the `skipif` litmus tests (`tests/test_ablation.py::test_litmus_reproduces_gate_a`).

Also queued (owner-gated / pending input):
- Ship-gate target calibration ? **#172** (owner decision; needs corpus growth).
- Tests reorganization ? blocked on the owner's grouping metric (provide it ? an agent will do it).
- Restore CI ? the Actions outage is account-level (billing/minutes), not a repo/config bug.

Leaving (bulky / M2 ? no start without owner go): ActionFormer (M0.4), event-index DB migration (M0.1), Gemini-silver purge (M0.3), cost-to-Gen-2 benchmark + ByteTrack deprecation, multisport, repo rename, PMF execution.

Where we are (one paragraph)
ADR-008 ratified ("repo split driven by productionization, not isolation"): training lives in-repo (`training/`)
behind a CI-enforced import wall; the featurizer is single-sourced (`backend/features/rally_seq.py`, train==serve);
the **Gen-0 harness ships** (`python -m training.gen0.harness --manifest output/human_lovo_manifest.json --floor
eval_baselines/heuristic_lovo_n6.json --version <semver>`) and produced artifact v0.1.0 (LOVO **0.626**, floor passed,
sign test 5/6 wins p=0.109 ? honestly not significant, ship=False). The **n=6 owned corpus is in the collector**
(262 rallies, Proprietary, commercial export = owned data only after the ShuttleSet reclassification). The **Gemini
battery is done** (see table) ? the moat is quantified.

The quality table that now drives strategy (IoU 0.5 vs human golden labels)
Path	Mahadevpura (clean-ish)	Kushagra (multi-court)
Heuristic (velocity windowing)	0.657	0.157
Gen-0 owned head (LOVO, n=6)	0.744	0.561
Two-stage WASB+Gemini refine	0.83	?
Gemini wrapper (`gemini-flash-latest`)	**0.93**	**0.66**

Reading: clean footage is commoditized (serve it with Gemini for cents). **Multi-court drops the commodity
wrapper to 0.66 ? that is the owned model's ship target** (its FPs are background-game

pickups + dead-time merges,
the exact contrast-metric confound). Gen-0 is 0.10 behind with only n=6 ? a corpus-growth-sized gap, not an
architecture-sized one. Two-stage refine = the cost-efficient serving path for LONG tapes (uploads candidate clips
only); wrapper 503-flakiness (observed all session) makes the provider-fallback registry load-bearing.
Baselines tracked in `eval_baselines/` (heuristic_lovo_n6.json, gemini_wrapper_2026-06-10.json).

Rally-detection quality track (design merged 2026-06-13, owner-gated; implementation in progress)
This is the track a desktop/local agent is slated to pick up. Design docs landed (PR #112 + session). All implementation is **owner-gated** (per explicit in the plan); each step = ONE PR off `master`.

Hardening loop (T1-T5 positive tests + audit + finale) COMPLETE (2026-06-14, PRs #156/#157/#160/#161/#163 + #165 + final records). See archived `docs/archives/CODE_AUDIT_AND_TEST_HARDENING-COMPLETED-2026-06-14.md` (Living Log with every pre/post/review) and `docs/archives/HARDENING_LOOP_HANDOFF-COMPLETED-2026-06-14.md` (full rules + STATE). Top-level now pointer stubs per #165 review. All 5 items + verification + archive done; clean slate.

- **Docs:** `docs/MULTI_SIGNAL_FUSION_PLAN.md` (shuttle + person + audio, one fusion layer) ·
  `docs/EXPERIMENT_HARNESS.md` (A/B + shadow + promotion gate, zero-regression) ·
  `docs/ROORA_LABELING_GUIDELINES.md` (v8) · `docs/HOW_RALLY_DETECTION_WORKS.md` (2-layer mental model).
- **Key finding (no re-investigation needed):** the owner's "held-shuttle counted as a rally" bug is **NOT a missing capability** ? `backend/pipeline/detectors/rally_gate.py` already implements the net-crossing **"Gate A"** (`apply_gate(..., min_crossings=2)`) that kills service-feed/held-shuttle windows, but it is **only called from eval drivers** (and there was no `min_crossings` knob in `IndexingConfig`). **Wiring Gate A into the live segmenter (now via fusion_hybrid) was the single cheapest, highest-confidence quality win.** (P2 FusionSegmenter specifics: registry `fusion_hybrid`, **default-OFF**, seeds from substrate, persists `net_crossings` + optional person features, optional served Gate A/B via config knobs; adversarially reviewed; suite green. See `tests/test_served_gate_a_regression.py` for the honest finding that on production windowing Gate A is neutral/negative in some cases, hence kept opt-in with knob for safety, and the leaderboard revert note.)
- **Current status (2026-06-14):** P1 (person cue extraction to reusable `PersonDetector`) + P2 (FusionSegmenter + Gate A signal + served handoff gating) + experiment harness P1 DONE (adversarial review + NO-REGRESSION; suite green). **P3 next (this PR + follow-ups):** learned fusion (person features ? harness `compare` LOVO lift on the 6 golden videos; eager-person-compute optimisation); then P4 audio via hit_detector.
- **Pick-up order (each = ONE PR off `master`, owner approval first, flag-gated default-OFF, promote only on measured LOVO):**
  1. (P2 first-step, largely done via fusion_hybrid) Wire Gate A into production served path + config knob (min_crossings); apply in hybrids at runtime. (Now available opt-in; served neutral on production windowing per litmus ? kept opt-in with knob for safety.)
  2. P3 first step (this work item): harness `compare` LOVO litmus for person-on variant (players_in_court + two_sided + colocation etc. from `fusion_features`) on the 6 golden; add the eager compute optimisation note/landing. Pure + harness (testable here).
  3. P3/P4 continuation: audio cue integration + full learned promotion if lift proven.
- **Discipline (from EXPERIMENT_HARNESS.md):** every new cue **flag-gated, default OFF**; promote to default **only on measured LOVO lift** through `run_regression.py` (exit 0/1/3); pinned `CONTROL` ? rollback = config flip. Harness re-uses ~80% existing code.
- ? Dev container no GPU/torch/cv2 ? Gate A, person cue logic, harness, and pure fusion paths are testable here; real-video / golden LOVO / ablation confirmation is owner-side (GPU/WSL machine).
- **External due-diligence review inputs (2026-06-17) ? PICK UP AFTER current ideas are exhausted; full details + citations in [`RALLY_QUALITY_RESEARCH.md`](RALLY_QUALITY_RESEARCH.md) §5.6.** An owner-commissioned 31-reference review of the quality report *validated* the methodology and surfaced these **future** levers (research inputs, not active work, not new claims; verify citations before any submission):
  1. **"First-mile" paper framing (highest-value):** stroke-classifiers (BST/DSTA) + the

big datasets (ShuttleSet ~36k strokes, Fine-Badminton ~27k actions) all **assume a pre-trimmed rally clip already exists** ? we solve the ignored prerequisite. Pitch the paper as *"the missing untrimmed-video preprocessing pipeline that lets BST/MonoTrack run on raw footage."*
    2. **Ground R2 in Action-Spotting / Precise-Event-Spotting** (SoccerNet/TenniSet/T-DEED) ? turns R2 into a defensible publishable contribution.
    3. **Add domain baselines for any submission:** MonoTrack + BST on ShuttleSet / Fine-Badminton (we have only heuristic + Gemini today).
    4. **Prefer TemporalMaxer over ActionFormer** for the M0.4 scorer swap (parameter-free max-pool, ~2.8× fewer GMACs, small-N robust) ? see "M0.4 next increment" below.
    5. **Court polygons ? pose ? the rally-END lever:** masking on-court players can resolve *occluded end-of-rally states* (the #1 remaining gap vs Gemini), not just spectator noise.
    6. **Name the eval?serve lesson "pipeline distribution skew"** ? a citable paper "lesson."

Discipline reminder: owner approval before starting any owner-gated item; one PR per step; babysit the PR (poll/fix/reply) until merge observed, *then* advance.

Prioritized next steps
1. **[owner, in progress] GROW THE GOLDEN CORPUS** ? the single highest-leverage move; every lever below feeds on it.
   Priority order for new videos: **(a) multi-court badminton** (the target distribution AND where the ship target
   lives), **(b) ?3 matches per new sport ? table tennis first** (WASB transfers at F1 0.909; pickleball labels are
   corpus-gold but not trainable until a clean MIT/Apache ball detector exists), (c) capture variety per
   `VIDEO_RECORDING_GUIDELINES.md`; **adult sessions only for the training pool** (consent guardrail). Per video:
   label ? `curate import-local --normalize --license Proprietary --fps <true fps>` ? re-run the Gen-0 harness.
   The paired sign test needs n: at n=10 with 9 wins, p?0.011 ? significance is reachable this month.
2. **[owner decision] Set the ship-gate target** ? now informed: floor = heuristic 0.480 (necessary), **multi-court
   reference = wrapper 0.66** (the number that makes the owned model worth serving), clean-footage ceiling = 0.93
   (not the target ? Gemini serves that tier). Suggested shape: "LOVO F1@tIoU0.5 ? 0.70 on multi-court folds with
   paired-sign p<0.05 vs heuristic" ? owner to ratify/adjust.
3. **M0.4 next increment ($0/local):** swap the Gen-0 LogReg for a lightweight temporal head (MIT, minutes on the
   2070S ? NOT the heavyweight Gen-2 stack) inside the existing harness; per-video threshold calibration (the
   LOVO-vs-oracle gap is visible per fold in the harness output). **Prefer TemporalMaxer** (parameter-free
   max-pool, ~2.8× fewer GMACs, small-N robust) over ActionFormer/ASFormer per the 2026-06-17 external review
   (`RALLY_QUALITY_RESEARCH.md` §5.6) ? better fit for the cheap/local/fast mandate.
4. **Multisport thread:** ingest Roora pickleball/TT CSVs (`import-local --normalize`); merged-prototype LOVO across
   badminton+TT once TT has ?3 matches; identify a clean pickleball ball detector. ? TT-on-WASB serving stays gated
   by C3 (provenance multiplies per sport).
5. **PMF validation (cheaper than any engineering month):** C13 academy interviews (+ the two added questions:
   "multi-court footage nobody watches?" / "pay per-match to index it?"); camera pilot at the warmest 2-3 academies
   (consent at capture; demo served via the Gemini path or human-polish ? the reel pattern exists:
   `output/MahadevpuraSingles_highlights.mp4`). **Field kit COMPLETE (2026-06-12, owner ratified the
   physical-visit approach; reframed product-feedback-first per owner spec):**
   `docs/PRODUCT_FIELD_ASSOCIATE_FLYER.md` (the recruiting flyer ? owner fills ?/contact placeholders and recruits) +

106   `docs/FIELD_VISIT_PLAYBOOK.md` (the hired associate's visit playbook: coach + student script, demo-after-Q6
107   rule, ground rules, Phase-2 record-and-show-back stretch goal) + `docs/FIELD_VISIT_ANSWER_SHEET.md` (per-visit
108   capture form incl. demo-reaction + student blocks) + `docs/PMF_FIELD_SIGNALS.md` (G/A/R anchors + **pre-registered
109   decision rules** ? PROCEED / PIVOT A-C / FALSIFIED thresholds fixed before the first visit).
110   6. **Housekeeping:** ~~rotate the Gemini API key~~ **DONE 2026-06-10** (owner-verified working; confirm old-key
111   deletion in AI Studio); **rename repo off "badminton"** (owner action,
112   in BACKLOG); collector=SoR confirmation; off-peak niceties: none pending ? the battery completed.
113
114   ## Pointers / artifacts
115   - Roadmap: `docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md` · Topology: ADR-008 (`docs/archives/decisions/DECISIONS.md`) ·
116   Study: `docs/archives/research/OWNED_MODEL_TRAINING_STUDY.md` · Quality history:
117   `docs/archives/quality-iterations/2026-06-multivideo-lovo/`.
118   - Harness: `training/gen0/harness.py` (one command ? train ? LOVO ? gate ? artifact bundle; weights gitignored,
119   manifest reviewable). Featurizer: `backend/features/rally_seq.py` (FEATURE_SCHEMA v1).
120   - Corpus: collector DB (system-of-record, 6 matches/262 rallies) · eval manifest `output/human_lovo_manifest.json`
121   (gitignored) · durable trajectories + Roora files: `C:\Users\avidu\Projects\Annotation Setup\Collect\Trajectories\` ·
122   labels: `...\Collect\Golden Labelled\`.
123   - Gemini ops learnings: serial uploads (`indexing.ai_max_workers=1`) + chunk re-encode
124   (`indexing.ai_chunk_scale_width=1280`) + 6-attempt exponential generate retry ? a cold/prepaid project burst-throttles
125   with a MISLEADING "credits depleted" 429; ramp gradually.
126   - ? No owned-model implementation beyond greenlit M0.x until owner greenlights the next milestone. Committed next
127   PLATFORM slice = async job model (single-tenant).
128

6. user (2026-06-19T15:46:39.399Z)

1   - ROORA_LABELING_GUIDELINES.md ? Latest (v8) vendor labeling guidelines mirrored from the collector's `docs/annotation/`, with the fusion-driven court-calibration / singles-doubles additions marked `? ADDED (fusion)`.
2   - RALLY_QUALITY_RESEARCH.md ? **? Foundation doc** for closing the local-vs-Gemini rally-detection gap: the measured gap + the code-grounded over-merge diagnosis ("shuttle moving ? rally"), an inventory of the eval/experiment harness + its gaps, the strengthened evaluation framework (segmental Edit/F1@k that make over-segmentation first-class, golden-corpus + active learning, Gemini-agreement as a *shadow* signal), an **adversarially-verified, cited external-research synthesis** (TAL/TAS boundary heads, racket-sports rally-state cues, small-data, eval rigor), a **prioritized cheap?durable roadmap with falsifiable success criteria**, and (§7) a **tracked project plan** ? work-item checklist, sequencing, the single quality **number** attributed to the effort, and a **definition of done** (project completes when all tiers land + the number is recorded). The plan that the QUALITY_ITERATIONS.md log executes against; ties together EXPERIMENT_HARNESS / MULTI_SIGNAL_FUSION_PLAN / ANALYZERS_AND_RECONCILERS / OWNED_MODEL_IMPLEMENTATION_PLAN.
3   - RALLY_DETECTION_GAP_CLOSURE.md ? **Tracked execution doc (2026-06-18)** for closing the remaining rally-detection accuracy gap **per video, before launch**, driven by the growing golden corpus. First ground-truth clip (GX010141) reframed it: boundaries are tight (R2/R3/R4 holds) ? the gap is **detection**: (G1) recall (local & Gemini both miss ~half on hard clips, and *different* rallies ? complementary), (G2) precision (local over-fires), (G3) **rally-END over-extension on the quick-return-after-point** (shuttle still moving ? needs a *player-state* signal). Levers (owner-gated): player-kinematics + shuttle-in-hand / sustained-invisibility end-rule with reverse-timeline backtrack, `rally_gate` Gate A wiring, complementary fusion. **Peer** of `RALLY_QUALITY_RESEARCH.md` (execution tracker, not research); §7 is the source of truth.
4   - RALLY_DETECTION_QUALITY_REPORT.md ? **Technical-

paper checkpoint (2026-06-17)** of rally-detection quality: the honest measured state (over-segmentation milestone tolerance-F1 0.091?0.323 at n=11, p=0.002; merge-rate 0.694?0.150; first real-footage ground-truth at-par with START at Gemini parity), the task-faithful R2 metric, a detailed Gemini comparison, next steps + expected wins, areas yet to be explored, and a publishability / path-to-submission verdict. Renders to a polished PDF; the externally-shareable "toward a paper" summary of `RALLY_QUALITY_RESEARCH.md` + `QUALITY_ITERATIONS.md`.
- EVALUATION.md ? How we measure and improve quality.
- EXPERIMENT_HARNESS.md ? Design (P1 SHIPPED) for A/B + shadow + SxS experimentation on rally-detection variants with **zero production regression** ? experiments are decoupled from deploy (new cues default OFF, promote only on measured LOVO lift, rollback = flip a flag). Composes the existing `backend/eval/` machinery (`calibration`, `regression`, `metrics`); implemented in `backend/eval/experiment.py`.
- QUALITY_ITERATIONS.md ? **Living log** of rally-detection quality iterations (hypothesis ? what shipped ? measured/expected effect ? lesson). The reverse-chronological record; terminal-state snapshots get archived under `archives/quality-iterations/`.
- ANALYZERS_AND_RECONCILERS.md ? Design (owner-gated, NOT started): a **signal-fusion framework** for rally detection ? producers emit confidence-scored **signals** at three temporal **scopes** (frame detectors / window analyzers / session analyzers), a learned **scorer** weights them, and an auditable **report-first reconciliation** pass lints the decision against the evidence. The spine: **no special-casing in the decision path** (signals?scorer, never `if/else`). Worked examples: a service-fault analyzer, a warm-up/practice span detector, and consuming a per-frame shuttle-state detector. A peer to `EXPERIMENT_HARNESS.md` and `MULTI_SIGNAL_FUSION_PLAN.md` (ADR-007 modular rally layer).
- PERSONALIZATION_PLAN.md ? Design (owner-adopted **hybrid**; Phase-0 landing) for the **per-user personalization feature on a generalization-first baseline**: a thin per-user *tuning* layer gated by the honest small-N statistics (a **min-N promotion floor** + a **structural-guard exemption**, never a service gate) + a **contribution flywheel** (free-tier videos pool into the owner-trained baseline; *"a tool that gets better the more you help it"*). Records the **locked owner decisions** (identity, storage, `min_n` semantics, Gate-A default, free-tier pooling). Built so far: per-user promotion gate (#141) + held-out-USER learning curve (#142).

- GOLDEN_REGRESSION_FIXTURES.md ? **Design (owner-gated, NOT started):** video-free, non-attributable regression fixtures so a clean clone runs the rally-seg suite with **no video/GPU/DB**. Two layers ? the post-detector freeze + **dual gates** (characterization + quality-floor) reusing the `backend/eval/` stack, and the **SEALED-FIXTURES** privacy filter (de-attribution-at-source, tiered public/key-gated, shuttle-only public tier). Carries a cited **IN/US/EU legal memo** (derived-data IP/privacy), a threat model, and a **§7 progress tracker + definition-of-done**. Extends GOLDEN_DATA_SHARING.md.

Product / platform plans
- DESKTOP_APP_PLAN.md ? **Tracked scoping doc (2026-06-18, owner-gated, `[design]` ? NOTHING built):** package the local pipeline as a **cross-platform desktop app** for non-technical players ? plug in the camera's USB/SD card, click once, get back **only the highlight reels**, fully on-device (no upload, no original-video bloat). The centerpiece is the **de-WSL native-compute port** (run WASB natively per-OS: Linux/Windows CUDA · macOS MPS/CPU ? drop WSL2), and its **make-or-break unknown**: is CPU acceptable on a typical *no-GPU* enthusiast laptop (WASB is ~3.4× real-time on a laptop 2070S)? §7 tracker = **P0 compute-feasibility spike ? P1 native detector ? P2 app shell ? P3 packaging/installers**. Realizes the **L0 fully-local** tier of VIDEO_LOCALITY_MODEL.md; is the **`LocalDesktop` ComputeBackend** of PLATFORM_ARCHITECTURE.md §3.2; the smaller/export-clean owned model (OWNED_MODEL_IMPLEMENTATION_PLAN.md) is one answer to the compute crux.

Practical / ongoing
- TRACKNET_WSL_SETUP.md
- Golden set docs (`GOLDEN_SET_*`)
- Evaluation harness and calibration drivers (in `backend/eval/`)
- GOLDEN_DATA_SHARING.md ? How the GPU box and CPU dev/agent sessions share golden artifacts (trajectory CSVs, `*_golden_features.json`) via a OneDrive folder; the GPU-extract ? CPU-litmus split. See **#175**.

Conventions & templates
- PROJECT_DOC_TEMPLATE.md ? Reusable skeleton for a **tracked project doc**: status header · decisions-locked · progress tracker (?/?/?) · definition-of-done · archived on completion. Standardizes the emergent pattern in `RALLY_QUALITY_RESEARCH.md` §7 and `GOLDEN_REGRESSION_FIXTURES.md`. Use for any substantial design/project work.

Historical documents
Completed plans, old research, ADRs, and checkpoints live in [`archives/`](archives/).

- CODE_AUDIT_AND_TEST_HARDENING.md ? **COMPLETED** (2026-06-14; T1?T5 + full loop + verification per #162). Living record of the June 2026 plumbing / contract / architectural audit, 626-test infrastructure, positive test plans for the 5 identified issues, multi-layer regression protocol (including ablation/experiment layer), and pre/post-change verification checklist. See `docs/archives/CODE_AUDIT_AND_TEST_HARDENING-COMPLETED-2026-06-14.md` and `docs/archives/checkpoints/2026-06-14-hardening-loop-complete.md` for the final archived state. (Top-level is now pointer stub per review; full content in archives/.)

See `archives/README.md` for the umbrella policy and `archives/past_projects/README.md` for classification of small tech debt / starter projects.

7. assistant (2026-06-19T15:46:42.845Z)

Now let me verify the seeds. SEED 1 says the dev container CAN run the full pipeline. Let me verify cv2/numpy/torch import and check the actual repo state, and look at the NEXT_STEPS.md statements about CPU-only.

8. user (2026-06-19T15:46:51.524Z)

```
3.13.13 (tags/v3.13.13:01104ce, Apr  7 2026, 19:25:48) [MSC v.1944 64 bit (AMD64)]
---numpy---
numpy 2.4.6
---cv2---
cv2 4.13.0
---torch---
torch 2.6.0+cu124 cuda True
```

9. assistant (2026-06-19T15:46:55.366Z)

SEED 1 confirmed: numpy, cv2, and torch all import fine, and `torch.cuda.is_available()` returns True (GPU present). This directly contradicts CLAUDE.md line 61-62 and several NEXT_STEPS.md claims. Let me verify the seed about local windowing being default-ON is reflected correctly, and check whether NEXT_STEPS / README make stale claims about it. Let me look at git history and the relevant docs more carefully.

10. user (2026-06-19T15:46:57.025Z)

```
ebe2e6d fix(gemini): surface failed chunks loudly (no more silent partial reels)
a3b4954 feat(runner): --min-rally-duration flag (surface the long-rally reel filter on the CLI) (#238)
a240221 fix(test): ablation litmus uses canonical experiment._is_fusion_schema (unbreaks the suite) (#237)
d749fb9 test(eval): re-ground served Gate-A litmus on the present (multicourt) golden set (#236)
ea6b2b8 docs(desktop): scope a cross-platform desktop app + the de-WSL native-compute port (#219)
20f8aa3 feat(eval): P7 long-rally (>5s) quality lens + reel knob (both default-OFF) (#220)
5f555cf refactor(eval): extract one shared golden-feature loader behind the three load_videos (#235)
5409fc4 refactor(api): extract job-worker drain/wake loop from main into backend.api.job_worker (#234)
5c46ebc refactor(eval): split real-fixture subsystem into golden_real_fixtures (pure move) (#233)
97fa187 refactor(db): route bool-returning single-statement writers through _execute_write (#232)
```

```
ae10447 refactor(eval): extract _build_preset(args, side) from rally_seg_eval.main() (#231)
1e324ea refactor(motion): extract detection + DB-populate halves from process_video (#230)
53b9cef docs(code-map): point UPLOAD endpoints at backend/api/uploads.py (post #228) (#229)
e67add1 refactor(api): extract chunked-upload island from main into backend.api.uploads router
(#228)
8092a39 refactor(segmenter): decompose yolo_hybrid process_video into phase helpers (#227)
```

11. user (2026-06-19T15:46:58.891Z)

```
15:## ? Next-session / GPU-box pickup (2026-06-15)
18:**Immediate next step ? the real P3 LOVO measurement, split GPU?CPU** (the dev box is CPU-
only: `torch+cpu`/`cv2` present but no GPU, and the corpus/videos/CSVs/labels are not here):
19:1. **[owner / GPU box]** `python -m backend.eval.fusion_golden --manifest
output/human_lovo_manifest.json --out-dir "%USERPROFILE%\OneDrive\rally-annotator\golden-
data\features\2026-06-15-n6"` ? produces `*_golden_features.json` for the 6 golden videos. (GPU
+ WSL + videos + trajectory CSVs + `.rallies.csv` labels are owner-side only.)
21:3. **[CPU dev/agent session]** `python -m backend.eval.fusion_compare --features-dir
<shared>\features\2026-06-15-n6 --record` + `python -m backend.eval.ablation --features-dir
<shared>\... --granularity group` ? real person-feature ?F1; copy the JSONs into `output/` to
also pass the `skipif` litmus tests (`tests/test_ablation.py::test_litmus_reproduces_gate_a`).
69:- ? Dev container no GPU/torch/cv2 ? Gate A, person cue logic, harness, and pure fusion paths
are testable here; real-video / golden LOVO / ablation confirmation is owner-side (GPU/WSL
machine).
```

12. assistant (2026-06-19T15:47:05.546Z)

```
SEED 1 is strongly confirmed. NEXT_STEPS.md has multiple stale "no GPU/torch/cv2" / "CPU-only" /
"torch+cpu" claims that directly contradict the environment (torch 2.6.0+cu124, cuda True, cv2
4.13.0, numpy 2.4.6).

Now let me verify the seed about local windowing default-ON. The seed says to flag any meta doc
calling local windowing not-started / default-OFF / scaffolding-only. Let me check what
NEXT_STEPS.md and README say about the quality track / Gate A / windowing status, and verify
against code. Note the seed's specific subject is the cap=6 + R4 milestone (max_window_duration,
window_inactivity_end, etc.), which I've confirmed default-ON in models.py.

Let me check the "implementation in progress" vs "NOT started" language and the served Gate A
status more carefully.
```

13. user (2026-06-19T15:47:07.590Z)

```
4:owner-gated ? see the dedicated section below, the desktop-agent pick-up). Prior: 2026-06-10
(audit?M0?Gemini-battery session; later same day the **UI-alpha track opened**:
23:**Also queued (owner-gated / pending input):**
54:## Rally-detection quality track (design merged 2026-06-13, owner-gated; implementation in
progress)
55:**This is the track a desktop/local agent is slated to pick up.** Design docs landed (PR #112
+ session). All implementation is **owner-gated** (per explicit in the plan); each step = ONE PR
off `master`.
62:[Omitted long matching line]
64:- **Pick-up order (each = ONE PR off `master`, owner approval first, flag-gated default-OFF,
promote only on measured LOVO):**
68:- **Discipline (from EXPERIMENT_HARNESS.md):** every new cue **flag-gated, default OFF**;
promote to default **only on measured LOVO lift** through `run_regression.py` (exit 0/1/3);
pinned `CONTROL` ? rollback = config flip. Harness re-uses ~80% existing code.
78:**Discipline reminder:** owner approval before starting any owner-gated item; one PR per
step; babysit the PR (poll/fix/reply) until merge observed, *then* advance.
81:1. **[owner, in progress] GROW THE GOLDEN CORPUS** ? the single highest-leverage move; every
lever below feeds on it.
```

14. user (2026-06-19T15:47:07.958Z)

```
1       # Rally-detection quality iterations ? living log
2
```

Purpose. A running, reverse-chronological log of how we evolve rally-detection quality: each iteration's hypothesis, what shipped, the measured (or expected) effect, and the lesson. This is the **live** continuation of the historical trail in [`archives/quality-iterations/`](archives/quality-iterations/README.md) ? per the archive policy (`CLAUDE.md`), **terminal-state** (DONE/REJECTED) work gets a dated subdir there; this file stays at `docs/` top level and keeps growing, then its entries graduate to the archive when they conclude.

Methodology (non-negotiable). Every quality change rides the experiment harness (EXPERIMENT_HARNESS.md): new cues land **flag-gated, default OFF**; they're graded **offline against golden labels** (per-video + LOVO-honest); promotion to default happens **only on measured lift** through the `run_regression.py` gate; **rollback = flip a config flag, never `git revert`.** Cues are designed in MULTI_SIGNAL_FUSION_PLAN.md; scoring in EVALUATION.md; the 2-layer model in HOW_RALLY_DETECTION_WORKS.md.

Golden corpus = the 6 labelled matches in `output/human_lovo_manifest.json` (owner-confirmed 2026-06-13: use all 6 ? `adarsh_avi_singles, kushagra_singles, largetest_doubles, gbaaddy, testlarge_short, mahadevpura_singles`). The model trained on these is what we run alongside the existing detector on NEW footage for the side-by-side review.

Baselines that drive strategy (IoU 0.5 vs human golden labels)

Path	Mahadevpura (clean-ish)	Kushagra (multi-court)
Heuristic (velocity windowing)	0.657	0.157
Gen-0 owned head (LOVO, n=6)	0.744	0.561
Two-stage WASB+Gemini refine	0.83	?
Gemini wrapper (`gemini-flash-latest`)	0.93	0.66

Reading: clean footage is commoditized (serve via Gemini); **multi-court drops the commodity wrapper to 0.66 ? that is the owned model's ship target.** Baselines tracked in `eval_baselines/`. The owner's observed false positive ? a player **holding/flicking** the shuttle between points counted as a rally ? is the precision bug the fusion gates target.

Current track (live) ? newest first

P7 ? Long-rally (>5s) quality lens (eval) + reel-selection knob ? MEASURED, default-OFF *(2026-06-18)*

Hypothesis (owner). Local OVER-FIRES, and its false positives are mostly SHORT (motion blips, background-court fragments on multicourt). Filtering the eval to **long rallies (>5s)** ? the eye-catching highlights that carry the product ? should strip most short FPs and reveal local's real long-rally quality, answering the North-star "do we beat Gemini on the single-match long-rally happy

48   path?". **Duration** (end?start), NOT shot-count, is the filter: golden `shots_count` is blank and the
49   no-AI path reports it "Unknown", whereas duration comes from the start/end labels we already trust.
50
51   **What shipped (both default-OFF, zero behaviour change at the defaults).**
52   - **Eval lens** ? `rally_seg_eval.py --min-duration <s>` filters BOTH predictions AND golden GTs to
53   rallies longer than `s` *before* matching, consistently across the tIoU and R2 tolerance paths (a
54   FILTER, not a new metric). Plus `--stratify` / `--long-tau` (default 5s): a `{all vs long} × {single
55   vs multicourt}` aggregate table. Court type = the committed `MULTICOURT_CLIPS` set (the manifest is
56   gitignored, so the set is the auditable source), overridable by a manifest `court_type` field; kept
57   in sync with [`GOLDEN_VIDEOS.md`](GOLDEN_VIDEOS.md).
58   - **Product knob** ? `stitching.min_rally_duration` (s, 0 = OFF) drops short rally pairs before the
59   reel compile in `tools/generate_highlights.py` ? a cleaner reel-selection filter than
60   `min_shot_count` (which leans on the "Unknown" shot_count).

15. assistant (2026-06-19T15:47:14.965Z)

Now let me look at the QUALITY_ITERATIONS R1 milestone entry to confirm the cap=6/R4 default-ON status is documented as shipped (this gives me the corroborating evidence that the windowing milestone is default-ON, not a design proposal). Let me also check whether the "Current state (June 2026)" / scaffolding language in CLAUDE.md is in my slice and whether it's stale. The seed is specifically about local windowing default-ON, so let me focus on whether any meta doc calls windowing not-started/default-OFF.

The CLAUDE.md "Current state (June 2026)" section says "Scaffolding complete... No platform/owned-model implementation has started" ? that's about the platform/owned-model tracks (owner-gated roadmap intent), which I must NOT relitigate. But I should check if CLAUDE.md anywhere calls the local windowing track not-started.

Let me look at the rally-detection quality track section in CLAUDE.md (lines 16-21) and check whether Gate A "not wired into production" is still accurate, and whether windowing is described.

16. user (2026-06-19T15:47:16.978Z)

19: "Gate A" **not wired into production** ? cheapest win) ? `docs/EXPERIMENT_HARNESS.md` (A/B + shadow
33: greenlit Phase-0 milestone (`OWNED_MODEL_IMPLEMENTATION_PLAN.md`). **Do not start

17. user (2026-06-19T15:47:18.665Z)

62:**Measured (n=15 golden, milestone cap6+R4, LOCAL vs golden; reproduce: `python -m backend.eval.rally_seg_eval --stratify`):**
87:   mahadevpura_1's 63s blob, GX010141 14.8s) and cap=6 truncates genuinely-long rallies so the strict
104:### R1 ? MILESTONE LAUNCHED (cap=6 + R4 flipped default-ON) ? re-verified at n=13 *(2026-06-18)*
105:The R1 milestone is **shipped** (PR #204): the W1 cap=6 + R4 inactivity-end defaults are now ON.
107:the **[R1 Milestone Launch Report](R1_MILESTONE_LAUNCH_REPORT.md)** (per the Launch-Report convention).
109:**Re-verified on the grown n=13 corpus** (the 2 new Boxhill clips GX010137/GX010141) ? the milestone
114:| baseline ? milestone tolF1 | ?+0.232, 10/0, p=0.002 | **?+0.222, 12/0/1, p=0.0005** |
120:milestone 0.474. Independently reproduced by a 2nd harness (#214 `lovo_resweep.py`, matches to 3 dp;
121:LOVO picks cap=6 **13/13** on tIoU) and survived a 5-lens adversarial workflow (every LOO

fold clears
129:The one clip the cap6+R4 milestone still fails is the **continuously-active blob** (mahadevpura_1:
146:**MEASURED (n=11, on the cap6+R4 milestone, R2 `tolF1`):**
148:| clip | milestone | + R5 | windows (milestone ? +R5) |
160:it on selectively for continuously-active footage. The R1 net over-seg guard confirms it: milestone?+R5 net
163:### R1 ? milestone evidence (cap=6 + R4) + the net over-seg promotion guard *(2026-06-17)*
164:The R1 milestone is the smallest safe local-windowing flip: **W1 bounded windowing cap=6 + the R4
166:the full ablation ladder baseline?milestone under the **R2** `tolF1` on **n=11** with honest paired
167:sign-tests, and (b) ships the **net over-seg promotion guard** that unblocks the milestone.
175:| + cap=6 (R3) | 0.276 [0.174, 0.380] | 0.407 | 5.7 s |
176:| **+ R4 inact_end=1.5 (milestone)** | **0.323 [0.210, 0.433]** | **0.440** | **3.8 s** |
179:- **baseline ? milestone:** ?`tolF1` **+0.232** [+0.153, +0.304], **sign 10/0/1, p=0.002** (the 1 tie is the
181:- **cap=12 ? cap=6** (R3 was means-only ? now confirmed): ?`tolF1` **+0.108** [+0.070, +0.154], **sign 10/0/1,
182: p=0.002**. **cap=6 beats cap=12, significantly.**
183:- **cap=6 ? +R4:** ?`tolF1` **+0.047** [+0.022, +0.072], **sign 8/0/3, p=0.008** (reproduces the R3/R4 entry).
189:and the raw merge **count** is non-monotonic. The milestone trips strict on 10/11 videos, **yet it strictly
205:**R1 verdict:** the milestone (cap=6 + R4) clears the refined (net-guard) promotion bar ? a significant `tolF1`
207:(`indexing.max_window_duration: 6`, `window_inactivity_end: 1.5`, `inactivity_rally_thresh: 0.20`,
216:- **R3 ? re-scope the cap (12 ? 6 s).** The 12 s cap was too coarse. Sweep (W1, no R4): `tolF1` cap=6 **0.273**
217: > cap=8 0.234 > cap=12 0.171. So R1's recommended cap is **6 s**, not 12. (Means-based lead across the
219:- **R4 ? rally-END inactivity trim** (`window_inactivity_end`, **default-OFF**). The measured gap is the
227: **MEASURED (n=11, on top of W1 cap=6; sweet spot ?_end-window=1.5 s, rally_thresh=0.20, min_density=0.30):**
229: | metric | cap=6 | + R4 | ? |
238: 0.276 ? +R4 **0.323**. Default-OFF; promote via config once R1's net-over-seg guard + corpus growth land.
247: milestone knobs OFF ? guards shipped behaviour) and **MILESTONE** (R3 cap=6 + R4 inactivity trim ? guards the
249: MILESTONE tolF1 **0.323** / f1@0.5 0.440 (matches the R3/R4 row above ? cross-checks the harness).
323:frame-aligned, added to `human_lovo_manifest.json`). **n=8 milestone numbers (LOVO, tIoU-F1):** baseline 0.232 ? W1
324:0.403 ? W1+W2 0.419 ? **W1+W2+hardcap 0.437** (merge_rate 0.634 ? 0.000); milestone ? baseline?W1+W2 = **+0.187 F1@0.5**.
325:All flags **default-OFF**; promotion to a shipped milestone is owner-gated and re-run at the ~16-clip corpus.
333:every piece ? `max_window_duration`. Still only cuts (never merges ? recall can't drop). Threaded through
351:tradeoff and the motivation for the **boundary head / rally-state-aware split** (M1/R1), the next lever).
353:`max_window_duration` should be a true cap ? NOT by a golden lift (there is none at n=6). Promote/flip with

[Showing results with pagination = limit: 40]

18. assistant (2026-06-19T15:47:26.957Z)

This is very informative. The QUALITY_ITERATIONS.md R1 entry (line 104-107) confirms the cap=6 + R4 milestone is **shipped** (PR #204), flipped default-ON. Note line 219 says "R4 ? rally-END inactivity trim... **default-OFF**" but that's in an older R1 entry (2026-06-17) that's now superseded by line 104's launched entry ? but QUALITY_ITERATIONS.md is NOT in my slice (my slice is CLAUDE.md, docs/README.md, docs/NEXT_STEPS.md).

Now, the key question for my slice: does CLAUDE.md, README.md, or NEXT_STEPS.md make a stale claim that local windowing is not-started / default-OFF / scaffolding-only?

CLAUDE.md line 16-21: the "Rally-detection quality track" is described as "**design merged 2026-06-13, owner-gated, NOT started**". But the cap=6+R4 windowing milestone HAS shipped (PR #204, default-ON in models.py). Let me check this carefully ? the CLAUDE.md line says the track is "NOT started" and Gate A is "not wired into production". Let me verify whether the windowing W1/R4 work is part of this "quality track" and whether the "NOT started" claim is now contradicted by shipped code.

Let me read the full CLAUDE.md quality-track bullet and check the R1 Launch Report and the served Gate A regression test.

19. user (2026-06-19T15:47:28.850Z)

104   ### R1 ? MILESTONE LAUNCHED (cap=6 + R4 flipped default-ON) ? re-verified at n=13 *(2026-06-18)*
105   The R1 milestone is **shipped** (PR #204): the W1 cap=6 + R4 inactivity-end defaults are now ON.
106   Full quality-impact table on both rulers, the over-seg guard, and the honest caveats are recorded in
107   the **[R1 Milestone Launch Report](R1_MILESTONE_LAUNCH_REPORT.md)** (per the Launch-Report convention).
108
109   **Re-verified on the grown n=13 corpus** (the 2 new Boxhill clips GX010137/GX010141) ? the milestone
110   holds and the headline is *stronger* than at n=11:
111
112 | transition | n=11 (filed) | **n=13 (launched)** |
113 |---|---|---|
114 | baseline ? milestone tolF1 | ?+0.232, 10/0, p=0.002 | **?+0.222, 12/0/1, p=0.0005** |
115 | cap12 ? cap6 tolF1 | ?+0.108, 10/0, p=0.002 | **?+0.110, 12/0/1, p=0.0005** |
116 | cap6 ? +R4 tolF1 | ?+0.047, 8/0, p=0.008 | ?+0.040, **9/1/3**, p=0.0215 |
117 | net over-seg guard | PASS (mr 0.694?0.150) | **PASS** (mr 0.651?0.161, 0 regressions) |
118
119   Both new clips improve (GX010137 0.448?0.685, GX010141 0.333?0.431); tIoU ladder baseline 0.236 ?
120   milestone 0.474. Independently reproduced by a 2nd harness (#214 `lovo_resweep.py`, matches to 3 dp;
121   LOVO picks cap=6 **13/13** on tIoU) and survived a 5-lens adversarial workflow (every LOO fold clears
122   the bar; worst-fold drop GX010128 still ?+0.202, p=0.0010 ? no single clip drives it). **Honest notes:**
123   W1 cap is the dominant lever / R4 is the marginal +0.040 increment (1 loss = GX010137, a tolerance
124   artifact); the gate is the NET rate-based guard (strict=False by design); GX010141 is the worst-behaved
125   clip (future over-merge target). The nightly self-flags STALE post-flip ? re-baseline once with
126   `--update-baseline`. W2 / hard-cap / R5 stay default-OFF.
127
128   ### R5 ? blob-fallback for the continuously-active blob (default-OFF, stratum tool) *(2026-06-17)*
129   The one clip the cap6+R4 milestone still fails is the **continuously-active blob** (mahadevpura_1:
130   ~174 s of play with no inactivity gap to split on). The W1 gap-split finds no gap; R4 finds no slow
131   tail; the result is one ~66 s mega-window that matches nothing (**tolF1 0.000**, the degenerate
132   0-match outcome [`REAL_FOOTAGE_VALIDATION_REPORT.md`](REAL_FOOTAGE_VALIDATION_REPORT.md) §5 R5 names).
133
134   **R5** (`window_blob_fallback`, default-OFF) is the last resort: **after** the gap-split and the R4

135   trim have each had their chance, any group still longer than `window_blob_factor` × the cap is a
136   genuine blob ? midpoint-chop it to ? the cap, **log the forced cut**, and **flag every resulting
137   window `forced_cut=True`** (surfacing the clip as one that needs rally-STATE cues ? net-crossings /
138   two-sided motion ? not a bigger cap). `_blob_fallback_chop` in `tracknet_runner`, wired through

20. user (2026-06-19T15:47:30.208Z)

ff51b37 chore(eval): re-baseline the nightly after the R1 milestone flip (#204) (#216)
67b6927 feat(config): R1 milestone ? flip windowing defaults ON (cap=6 + R4), re-verified at n=13 (#204)
f5f339a fix(eval): fusion loaders skip windowing-only files on a mixed golden corpus (#210)
55e4c89 docs(archive): LOVO threshold re-sweep at n=13 ? milestone thresholds hold (#214)
8c68143 feat(eval): R1 ? net over-seg promotion guard unblocks the cap6+R4 milestone (#201)
bb4d6e2 feat(eval): Tier 1 (W1.5) ? hard-cap fallback enforces the window cap on continuously-active trajectories (#195)
a3b209a feat(windowing): Tier 1 (W1) ? bounded windowing fixes the over-merge (?F1 +0.139, 6/0) (#187)
ebdf331 fix: CI for PR171 - wrap long docstring lines (E501), use cast for json.load mypy compat, ensure files end with newline
23ad934 fix: CI for PR170 - wrap long docstring line (ruff E501) in gate_verdict
b649f1c address review on #169: revert frozen archive edit (should-fix); trim handoff to short pointer per #165 intent (record moved to plan); add Transition Record section; unbundle catalog items 3/6/10 + clarify P3 'shipped' state with investigation (no fusion_compare.py etc. present; general harness + P2 are); add M0 Quick Wins (the 4 in-container items per course-correction); re-sequence milestones (M0 quick wins first + owner-led parallel, heavy last); update risk table + sequencing text. Thorough point-by-point in PR comment.
1b2450a docs: finalize handoff/audit after #166 merge + introduce post-166 work items plan with risk assessment, milestones and sequencing (review first per owner)
4251c9c fix(eval): close the eval?serve windowing gap ? single-source SERVED preset + served-regression promotion guard (#154)
53bc743 docs(quality): served-side Gate-A check ? flip's offline win doesn't transfer to production windowing (#152)
7b9dad2 fix(tests): cross-platform support for Windows-style paths in eval batch resolver and tracknet/wasb basename logic
8f4c564 Add real-time progress logging for MD5 hashing, add processing milestones tracking in Gemini segmenter, and bypass heavy OpenCV validations on large files

21. assistant (2026-06-19T15:47:40.639Z)

Confirmed: PR #204 (commit 67b6927) "feat(config): R1 milestone ? flip windowing defaults ON (cap=6 + R4)". The local windowing milestone is shipped/default-ON.

Now I need to assess my slice carefully:

1. **CLAUDE.md line 16**: "Rally-detection quality track (design merged 2026-06-13, owner-gated, **NOT started**)". This is now contradicted ? the W1/R4 windowing milestone (part of this quality track, RALLY_QUALITY_RESEARCH W1/R-series) HAS started AND shipped (PR #204 default-ON). However, I need to be careful: this CLAUDE.md bullet specifically frames the track as the fusion/Gate-A/experiment-harness sequence. The bullet also says Gate A is "not wired into production." Let me check the precise scope. The phrase "NOT started" is a status claim that contradicts shipped code. This is a factual staleness.

2. **CLAUDE.md line 61-62**: "This dev container lacks GPU / torch / cv2 / numpy ? the full pipeline can't run here." SEED 1 ? confirmed stale (torch+cuda, cv2, numpy all present).

3. **NEXT_STEPS.md line 18**: "the dev box is CPU-only: `torch+cpu`/`cv2` present but no GPU" ? stale (GPU present, torch+cu124).

4. **NEXT_STEPS.md line 69**: "Dev container no GPU/torch/cv2" ? stale.

Let me verify whether the CLAUDE.md "NOT started" is truly about the windowing track or strictly about fusion/Gate-A. The bullet enumerates: HOW_RALLY_DETECTION_WORKS ? MULTI_SIGNAL_FUSION_PLAN (Gate A) ? EXPERIMENT_HARNESS ? ROORA. This is the fusion track specifically. The windowing (W1/R4) comes from RALLY_QUALITY_RESEARCH.md, which isn't in this bullet's list. So "NOT started" here is scoped to the fusion track. Let me check whether the fusion track has started (FusionSegmenter built, experiment harness built).

22. user (2026-06-19T15:47:41.298Z)

```
backend\pipeline\detectors\fusion_features.py
backend\eval\fusion_golden.py
backend\pipeline\segmenters\fusion_hybrid.py
backend\eval\fusion_audio.py
backend\eval\fusion_compare.py
```

23. user (2026-06-19T15:47:41.766Z)

```
backend\eval\experiment.py
```

24. user (2026-06-19T15:47:43.535Z)

```
__init__.py
__pycache__
base.py
fusion_hybrid.py
gemini.py
motion.py
tracknet_hybrid.py
trajectory_hybrid.py
wasb_hybrid.py
yolo_hybrid.py
---detectors---
__init__.py
__pycache__
audio_features.py
base.py
fusion_features.py
person_detector.py
rally_gate.py
tracknet_runner.py
wasb_infer.py
wasb_runner.py
```

25. assistant (2026-06-19T15:47:51.678Z)

The fusion track HAS started and shipped substantial code: `fusion_hybrid.py` segmenter, `rally_gate.py` (Gate A), `person_detector.py`, `experiment.py`, `fusion_compare.py`, `ablation.py`, `FusionConfig` in models.py. So CLAUDE.md line 16's "NOT started" for the rally-detection quality track is contradicted by code.

However ? I must be careful and surgical here. The CLAUDE.md "Current state" bullet (line 26-34) and the quality-track bullet (16-21) carry roadmap intent. The task says "Only fix factual contradictions, not roadmap intent." The "(design merged 2026-06-13, owner-gated, NOT started)" is a factual STATUS claim. Let me verify the served Gate A "not wired into production" claim precisely ? is Gate A now wired into the served path?

The FusionConfig docstring (models.py lines 84-114) says the fusion segmenter "does NOT gate the served handoff unless a threshold is raised (`min_crossings`/`min_players` default 0 = OFF)". So served Gate A is BUILT but default-OFF. The CLAUDE.md says Gate A is "**not wired into production**" ? but it IS now wired (available via `fusion_hybrid` + `min_crossings` config knob), just default-OFF. Let me check the served Gate A regression test referenced in NEXT_STEPS to confirm the precise state.

26. user (2026-06-19T15:47:54.127Z)

950106b fix(fusion): W2 net-crossings gate is do-no-harm ? never empty a non-empty video (#188 follow-up) (#190)
5cc0ea4 test + docs: hardening loop completion record + rally quality track P3 first step (harness LOVO for person features); identify next candidates (per NEXT_STEPS). Small landing comment in fusion_hybrid + explicit list in NEXT_STEPS. Syntax green. Pre: container baseline (ast + obsreport note). Branched, one item, will babysit. Verification: owner re-run full + real video.
cbf2236 feat(segmenters): FusionSegmenter ? shuttle + net-crossing + person-cue fusion (P2, default-OFF) (#117)
8615534 refactor: adopt approved layered directory structure
---when added---
cbf2236 feat(segmenters): FusionSegmenter ? shuttle + net-crossing + person-cue fusion (P2, default-OFF) (#117)

27. user (2026-06-19T15:47:55.679Z)

```
16:- ``_select_for_handoff`` ? optional served Gate A/B: when ``indexing.fusion.min_crossings``
39:class FusionSegmenter(TrajectoryHybridSegmenter):
84:                                        min_crossings=0,
estimate=self._axis_estimate(points))
150:        where no segment clears ``min_crossings``), the gate falls back to the full ungated
158:        min_crossings = int(fcfg.get("min_crossings", 0))
164:            if min_crossings > 0 and st.get("net_crossings", 0.0) < min_crossings:
176:segmenter_registry.register(FusionSegmenter)
```

28. assistant (2026-06-19T15:48:10.343Z)

Confirmed. The `FusionSegmenter` is registered (`fusion_hybrid`), Gate A is available via the served path (`_select_for_handoff` applies it when `min_crossings > 0`), but it's default-OFF. So Gate A IS now "wired" into the segmenter, but default-OFF (not gating in production).

Now, the precise CLAUDE.md claim on line 19 reads: the held-shuttle bug is the already-built `rally_gate.py` "Gate A" **not wired into production**. At the time CLAUDE.md was written, Gate A was only called from eval drivers. Now (PR #117) it's wired into the `fusion_hybrid` served segmenter via the `min_crossings` knob ? but that segmenter is not the default (`default_segmenter = "yolo_hybrid"`) and the knob defaults to 0=OFF. So "not wired into production" is now partially stale: the wiring EXISTS (built, opt-in, default-OFF), it just isn't enabled by default.

This is nuanced. The seed for my slice focuses on the windowing milestone, and explicitly says "Only fix factual contradictions, not roadmap intent." The "Gate A not wired into production" claim is borderline ? the wiring code now exists but is default-OFF/not-the-default-segmenter, so "not wired into production" is arguably still accurate in the production-default sense. I'll be conservative and NOT flag the Gate A wiring line, since the default segmenter is still `yolo_hybrid` and Gate A is default-OFF ? calling it "wired into production" would over-claim, and "not wired into production (by default)" is defensible. The strongest, cleanest contradiction is "NOT started" for the whole quality track.

Let me re-read CLAUDE.md line 16 precisely. It says the track's design "merged 2026-06-13, owner-gated, NOT started." But fusion_hybrid (#117), rally_gate wiring, experiment harness, AND the windowing milestone (#204, default-ON) have all shipped. "NOT started" is factually false. The minimal surgical fix: the track HAS started (multiple PRs shipped, windowing milestone default-ON). But I must preserve voice and not relitigate owner-gating intent.

Let me check the CLAUDE.md "Current state" para once more and decide. I'll also double check the NEXT_STEPS "implementation in progress" vs other "NOT started" language to avoid double-flagging where the doc is already correct. NEXT_STEPS line 54 already says "implementation in progress" ? so NEXT_STEPS is internally inconsistent but at least line 54 is current. The stale items in NEXT_STEPS are the GPU/CPU environment claims.

Let me verify the exact uniqueness of the strings I'll target, and confirm there's not another "lacks GPU" type claim I'm missing in README.

29. user (2026-06-19T15:48:12.236Z)

11:[Omitted long matching line]
14:[Omitted long matching line]
20:- GOLDEN_DATA_SHARING.md ? How the GPU box and CPU dev/agent sessions share golden artifacts (trajectory CSVs, `*_golden_features.json`) via a OneDrive folder; the GPU-extract ? CPU-litmus split. See **#175**.

30. user (2026-06-19T15:48:13.368Z)

8:[Omitted long matching line]
11:[Omitted long matching line]

31. assistant (2026-06-19T15:48:20.215Z)

README line 20's "GPU box and CPU dev/agent sessions" / "GPU-extract ? CPU-litmus split" describes the GOLDEN_DATA_SHARING workflow convention (a documented two-machine split convention), not a hard claim that THIS environment can't run GPU. That's a convention/workflow description, and it's tied to where the corpus/videos live (owner-side), not purely GPU availability. I'll be conservative and not flag README line 20 ? it describes an artifact-sharing convention rather than asserting "no GPU here." The README "design (owner-gated, NOT started)" tags on lines 8/11/14 are for individual design docs (ANALYZERS_AND_RECONCILERS, GOLDEN_REGRESSION_FIXTURES, DESKTOP_APP_PLAN) ? let me verify those are genuinely not-built before deciding, since a "[design] NOT started" tag is stale only if the thing is actually built.

32. user (2026-06-19T15:48:21.130Z)

8   - ANALYZERS_AND_RECONCILERS.md ? Design (owner-gated, NOT started): a **signal-fusion framework** for rally detection ? producers emit confidence-scored **signals** at three temporal **scopes** (frame detectors / window analyzers / session analyzers), a learned **scorer** weights them, and an auditable **report-first reconciliation** pass lints the decision against the evidence. The spine: **no special-casing in the decision path** (signals?scorer, never `if/else`). Worked examples: a service-fault analyzer, a warm-up/practice span detector, and consuming a per-frame shuttle-state detector. A peer to `EXPERIMENT_HARNESS.md` and `MULTI_SIGNAL_FUSION_PLAN.md` (ADR-007 modular rally layer).
9   - PERSONALIZATION_PLAN.md ? Design (owner-adopted **hybrid**; Phase-0 landing) for the **per-user personalization feature on a generalization-first baseline**: a thin per-user *tuning* layer gated by the honest small-N statistics (a **min-N promotion floor** + a **structural-guard exemption**, never a service gate) + a **contribution flywheel** (free-tier videos pool into the owner-trained baseline; *"a tool that gets better the more you help it"*). Records the **locked owner decisions** (identity, storage, `min_n` semantics, Gate-A default, free-tier pooling). Built so far: per-user promotion gate (#141) + held-out-USER learning curve (#142).
10
11   - GOLDEN_REGRESSION_FIXTURES.md ? **Design (owner-gated, NOT started):** video-free, non-attributable regression fixtures so a clean clone runs the rally-seg suite with **no video/GPU/DB**. Two layers ? the post-detector freeze + **dual gates** (characterization + quality-floor) reusing the `backend/eval/` stack, and the **SEALED-FIXTURES** privacy filter (de-attribution-at-source, tiered public/key-gated, shuttle-only public tier). Carries a cited **IN/US/EU legal memo** (derived-data IP/privacy), a threat model, and a **§7 progress tracker + definition-of-done**. Extends GOLDEN_DATA_SHARING.md.
12
13   ## Product / platform plans
14   - DESKTOP_APP_PLAN.md ? **Tracked scoping doc (2026-06-18, owner-gated, `[design]` ? NOTHING built):** package the local pipeline as a **cross-platform desktop app** for non-technical players ? plug in the camera's USB/SD card, click once, get back **only the highlight reels**, fully on-device (no upload, no original-video bloat). The centerpiece is the **de-WSL native-compute port** (run WASB natively per-OS: Linux/Windows CUDA · macOS MPS/CPU ? drop WSL2), and its **make-or-break unknown**: is CPU acceptable on a typical *no-GPU* enthusiast laptop (WASB is ~3.4× real-time on a laptop 2070S)? §7 tracker = **P0 compute-feasibility spike ? P1 native detector ? P2 app shell ? P3 packaging/installers**. Realizes the **L0 fully-local** tier of VIDEO_LOCALITY_MODEL.md; is the **`LocalDesktop` ComputeBackend** of PLATFORM_ARCHITECTURE.md §3.2; the smaller/export-clean owned model

(OWNED_MODEL_IMPLEMENTATION_PLAN.md) is one answer to the compute crux.
15

33. user (2026-06-19T15:48:21.295Z)

No files found

34. assistant (2026-06-19T15:48:31.842Z)

ANALYZERS_AND_RECONCILERS (no analyzer/reconciler .py exists) and GOLDEN_REGRESSION_FIXTURES (a fixtures design) and DESKTOP_APP_PLAN are genuinely not-built design docs ? their "NOT started" tags are accurate, not stale. Good, I won't flag those.

Now let me make my final determination on the CLAUDE.md "Rally-detection quality track ... NOT started" claim. The fusion track (fusion_hybrid #117, rally_gate wiring, experiment.py harness) AND the windowing milestone (#204 default-ON) have shipped. So "NOT started" is factually contradicted. But I need to be precise about minimal surgical editing and preserving voice/owner-gating intent. The cleanest correction: the track is owner-gated but implementation has started (multiple PRs shipped, windowing milestone default-ON) ? matching NEXT_STEPS.md line 54 which already says "implementation in progress." Let me make the CLAUDE.md status consistent with that.

Let me also reconsider the "Gate A not wired into production" sub-claim within the same CLAUDE.md bullet. Since I'm editing that bullet anyway, but the rule is minimal surgical edits with unique stale_text ? I'll keep the two as separate, narrowly-scoped findings. Actually, for the CLAUDE.md bullet, the most defensible single contradiction is the "NOT started" parenthetical. The "Gate A not wired into production" is genuinely nuanced (default segmenter is still yolo_hybrid, Gate A default-OFF) ? I'll leave that phrase alone to avoid over-claiming. I'll narrowly fix only "NOT started."

Let me verify the exact text uniqueness for each finding before writing them.

35. user (2026-06-19T15:48:37.743Z)

```
=== CLAUDE.md NOT started count ===
1
=== CLAUDE.md lacks GPU count ===
1
=== NEXT_STEPS CPU-only count ===
1
=== NEXT_STEPS no GPU/torch/cv2 count ===
1
```

36. assistant (2026-06-19T15:48:47.661Z)

All four target strings are unique. Now let me finalize my findings. Let me reconsider the NEXT_STEPS.md line 18 carefully ? it says "the dev box is CPU-only: `torch+cpu`/`cv2` present but no GPU, and the corpus/videos/CSVs/labels are not here." The corpus-not-here part may still be true, but "CPU-only / torch+cpu / no GPU" is contradicted (torch 2.6.0+cu124, cuda True). The minimal surgical fix corrects only the GPU/torch claim while preserving the corpus-location claim.

Similarly NEXT_STEPS line 69: "Dev container no GPU/torch/cv2 ? Gate A, person cue logic, harness, and pure fusion paths are testable here; real-video / golden LOVO / ablation confirmation is owner-side (GPU/WSL machine)." The "no GPU/torch/cv2" is stale; the rest (corpus/video/golden owner-side) is a separate (still-plausible) claim.

Let me now write the structured output with my verified findings. I have four high-confidence factual contradictions:

1. CLAUDE.md line 61-62 ? "lacks GPU / torch / cv2 / numpy" (SEED 1)
2. NEXT_STEPS.md line 18 ? "the dev box is CPU-only: torch+cpu/cv2 present but no GPU"
3. NEXT_STEPS.md line 69 ? "Dev container no GPU/torch/cv2"

4. CLAUDE.md line 16 ? quality track "NOT started" (contradicted by shipped fusion track + #204 windowing default-ON)

For #4, I'll keep the edit minimal: change "NOT started" to reflect that implementation has started (consistent with NEXT_STEPS line 54 "implementation in progress"), preserving the "owner-gated" framing (roadmap intent untouched).

37. user (2026-06-19T15:49:02.775Z)

Structured output provided successfully

38. assistant (2026-06-19T15:49:15.781Z)

I've completed the staleness audit of my slice (CLAUDE.md, docs/README.md, docs/NEXT_STEPS.md) and returned four verified findings via StructuredOutput.

Summary of what I verified and found:

Confirmed stale (4 findings):

1. **CLAUDE.md:61-62** ? "This dev container lacks GPU / torch / cv2 / numpy ? the full pipeline can't run here." Directly contradicted: `torch 2.6.0+cu124` with `cuda.is_available()==True`, `cv2 4.13.0`, `numpy 2.4.6` all import here. (SEED 1 confirmed.)

2. **docs/NEXT_STEPS.md:18** ? "the dev box is CPU-only: `torch+cpu`/`cv2` present but no GPU." Same contradiction; corrected while preserving the still-true "corpus/videos not here" part.

3. **docs/NEXT_STEPS.md:69** ? "Dev container no GPU/torch/cv2." Same contradiction.

4. **CLAUDE.md:16** ? Rally-detection quality track tagged "NOT started." Contradicted by shipped code: `fusion_hybrid.py` (#117, registered segmenter), `rally_gate.py` Gate A wired into the served handoff, `backend/eval/experiment.py` harness, and the W1 cap=6 + R4 windowing milestone flipped **default-ON** in PR #204 (`backend/config/models.py`: `max_window_duration=6.0`, `window_inactivity_end=1.5`, `inactivity_rally_thresh=0.20`, `inactivity_min_density=0.30`). NEXT_STEPS.md:54 already says "implementation in progress" for the same track. (SEED 2 corroborated.)

Checked and deliberately NOT flagged (not stale / out of scope):
- CLAUDE.md "Gate A not wired into production" (line 19) ? wiring now exists via `fusion_hybrid` + `min_crossings`, but default segmenter is still `yolo_hybrid` and the knob defaults to 0=OFF, so "not wired into production (by default)" remains defensible; calling it production-wired would over-claim.
- CLAUDE.md "Current state / Scaffolding complete / No platform/owned-model implementation has started" ? owner-gated roadmap intent about the platform/owned-model tracks, not the local windowing track; out of scope per instructions.
- README.md "[design] NOT started" tags for ANALYZERS_AND_RECONCILERS (no analyzer/reconciler `.py` exists), GOLDEN_REGRESSION_FIXTURES, and DESKTOP_APP_PLAN ? genuinely unbuilt, tags accurate.
- README.md:20 GOLDEN_DATA_SHARING "GPU box ? CPU dev/agent" split ? describes an artifact-sharing convention tied to corpus location, not a hard "no GPU here" assertion.

The corrections preserve each doc's voice and owner-gating framing, touching only the factual GPU/environment and track-status contradictions.